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Method for judging parking status based on yolov2 target detection algorithm

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Abstract

In order to effectively utilize the public parking space resources on the roadside and realize smart transportation more comprehensively, this paper proposes a method based on machine vision to distinguish parking spaces. This method uses the yolov2 target detection algorithm, uses the camera to collect the image information of the parking spaces and vehicles, uses the yolov2 target detection algorithm to build the vehicle and parking space contour detection model, and obtains the parking space and vehicle center coordinate information, and calculates the vehicle and parking space contours through the parking space discrimination algorithm. To determine whether there is a vehicle parked in the parking space, determine the status of the parking space (occupied/vacant). The training accuracy rate is 99.1%. The results show that: the parking space recognition method has the advantages of simple structure, high recognition accuracy, not easily affected by external conditions, and wide application scenarios.

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Keywords: parking status identification; yolo; machine vision; smart transportation

1. Main text

With the increasing number of private cars, parking spaces are becoming more and more tense, and there is no place to park, which not only causes unnecessary public security problems and economic losses, but also causes the car owners to park vehicles at will, resulting in traffic congestion and road safety hazards. Moreover, there is a lack of supervision of parking spaces on the road, and the parking lot is only marked with or without parking spaces or the number of parking spaces, resulting in lack of guidance and unable to meet the needs of car owners. Therefore, how to make parking convenient and convenient has become a problem that must be faced in urban construction. The system developed in this project enables car owners to accurately understand parking space information on their mobile phones in real time and choose the best, which greatly facilitates car owners, reduces environmental pollution, and reduces potential safety hazards.

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There have been many researches and practices for smart parking systems. Many smart systems have been designed based on the current Internet of Things, visual recognition, image processing and other technologies and various algorithms. Among them, the mainstream ones are:

- Apply computer vision and image processing technology to achieve precise guidance of parking spaces. Secondly, the application of graph theory knowledge for abstract modeling, and finally based on the traditional Dijkstra algorithm, an optimal path planning guidance model is designed. ^[1]
- Intelligent parking guidance system based on ZigBee wireless communication technology. The system uses magnetoresistive sensors as parking space detection sensors, combined with parking space detection algorithms to determine the parking space status, and realizes parking space detection, data collection, data transmission, data processing, parking space guidance and real-time information display, Storage and other functions. ^[2]

Although the above two mainstream methods can achieve basic functions, their shortcomings are also very obvious: the implementation method is more complicated, and if the application is adopted, it needs to spend a lot of time on infrastructure construction, high cost, and time-consuming.

Therefore, this article proposes a simpler way, using only the pictures collected by surveillance or high-position cameras, and using the calculated yolov2 target detection algorithm to mark the range of empty parking spaces. When a car enters the screen, the outline of the vehicle and its center coordinates will be calculated, and the information will be stored. The coincidence rate of the vehicle and the outline of the parking space is calculated by the parking space discrimination algorithm to determine whether the parking space is parked, that is, whether it is in an idle state. If the coverage rate is higher than the set threshold, it is judged that the parking space is occupied; otherwise, it is idle. After testing, the training accuracy of the model can reach more than 99%. The method can be used with the mobile terminal for feedback at the same time, has high practicability, and has a wide range of application scenarios, and realizes the further construction and development of smart transportation.

2. Data Sources

2.1. Get The Data Set

Obtaining the data set is a complicated process. By using drones when the light is good during the day, the pictures of parking spaces on the roadside are collected and passed into the model for training to form a database of vacant parking spaces. In practical applications, the location of the parking space is fixed at all times, so only the model trained from the pictures collected during the day when the light is good can be used, and there is no need to consider the error of the camera to collect the marking information of the parking space at night, rainy and other bad weather.

In order to establish the vehicle contour recognition model and facilitate the practical application test, the drone was used to collect more than 600 pictures in the moving and stationary process of the vehicle during the day and night, and the ratio of 8:2 was divided into the training set and transferred to the model. Carry out training and test sets for testing and verifying results.

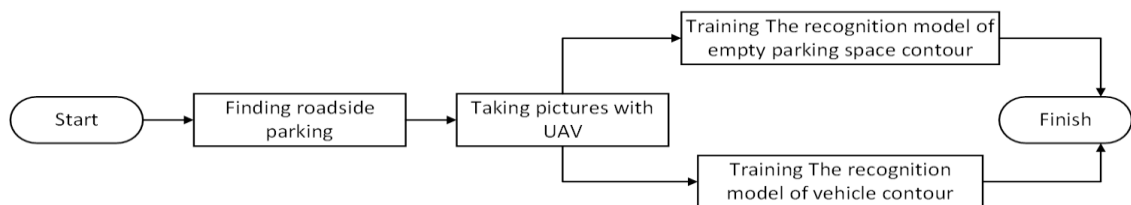


Fig. 1. Flowchart for data sets

2.2. Data Augmentation

In order to improve the precision and accuracy of the data set, we enhance the data by flipping the picture of the data set, zooming in on the partial data set, and simulating a variety of complex environments.

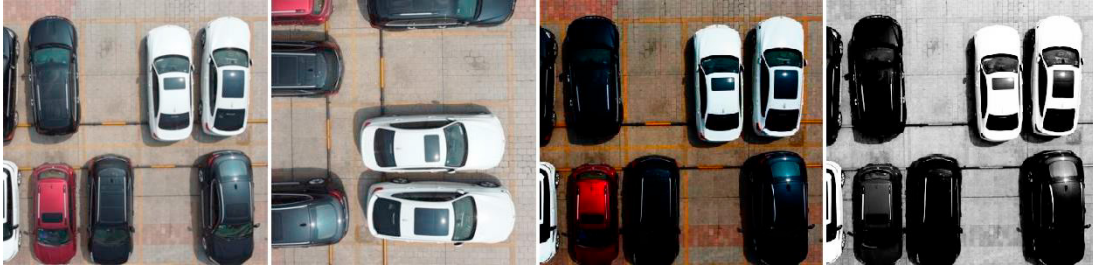


Fig. 2. concrete operations

3. Model establishment and data analysis and calculation

The main ideas of this algorithm are as follows:

After the data set formed by taking pictures, the yolov2 target detection algorithm is used to identify the parking space and the vehicle center coordinate information, and the parking space discrimination algorithm is used for comparison to obtain the idle state of the parking space.

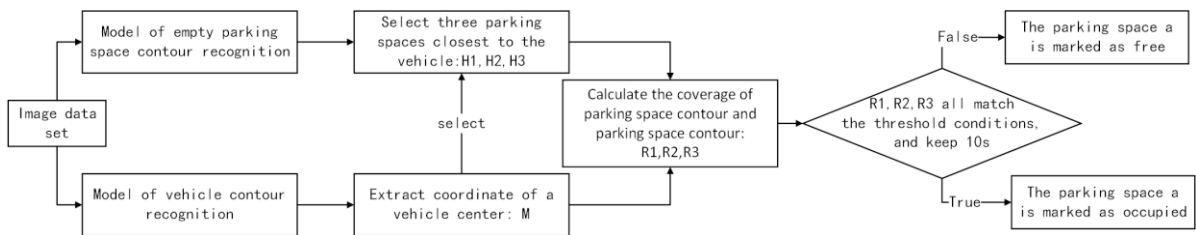
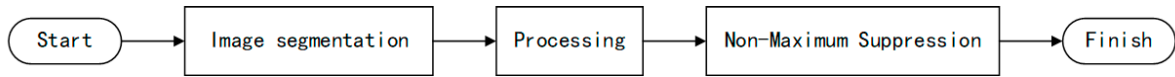


Fig. 3. concrete operations

3.1. The core idea of Yolo target detection algorithm

- Divide the input image into $S \times S$ gridcells (For example, it can be divided into 7×7 or 9×9 cells, and there is only one center cell.); if the center of an object is located in this gridcell, then this gridcell is responsible for detecting the object method, that is, considering which grid the center of the frame is in.
- Each gridcell needs to predict B boundingboxes coordinate information (x, y, w, h) and the confidence of these boundingboxes. The confidence level reflects the probability that the model contains objects in the boundingbox of this prediction.
- Each gridcell also needs to predict the probability of C categories to determine which category the current cell belongs to
- The class information of each gridcell is multiplied by the confidence information predicted by each boundingbox, and then the class-specific confidencescore of each boundingbox can be obtained.^[3]

Fig. 4. YOLO Core Block of Algorithm^[4]

The YOLO V2 algorithm has been improved in terms of accuracy, speed, and loss function. The effect has increased from 63.4% mAP of YOLOv1 to 78.6% mAP of YOLOv2, which is basically the same as FasterR-CNN and SSD.^[5]

3.2. Parking status judgment algorithm

- The parking status is divided into two types: occupied (flag=1) and idle (flag=0).
- (X_0, Y_0) is the coordinates of the center point of the current vehicle, respectively, and the C_0, K_0 is the length and width of the current vehicle identification frame
- (M_i, N_i) Are the coordinates of the center point of the current i-th parking space, and the C_i, K_i are the length and width of the current parking space identification frame respectively
- The R_i is the coverage rate between the rectangular frame of the vehicle and the rectangular frame of the i-th parking space

3.2.1. Choose The Nearest Parking Space

Based on a large number of data studies, it is found that an ordinary car has an impact on at most three parking spaces. H the distance of the current vehicle corresponding to different parking spaces:

$$H_i = \sqrt{(X_0 - M_i)^2 + (Y_0 - N_i)^2} \quad (1)$$

Record the initial i values of the first three bits after arranging the H_i from small to large.

For example, if the existing five parking spaces are sorted from small to large H the value is $[H_3, H_5, H_1, H_2, H_4]$, we record the center coordinates and length and width of parking spaces No. 3, No. 5, and No. 1

3.2.2. Determine the direction of the vehicle relative to the parking space and calculate the coverage

The coincidence rate of the vehicle corresponding to the last three parking spaces is calculated respectively. (The nearest parking space coincidence rate is R_1 , the second nearest parking space coincidence rate is R_2 , the third nearest parking space coincidence rate is R_3)

R formula is as follows (black box is parking space identification box, red box is vehicle identification box):

$$Case1: \begin{cases} X_0 - M_i > 0 \\ Y_0 - N_i > 0 \end{cases}$$

$$Case2: \begin{cases} X_0 - M_i > 0 \\ Y_0 - N_i < 0 \end{cases}$$

$$S_i = C_i * K_i \quad (1-1)$$

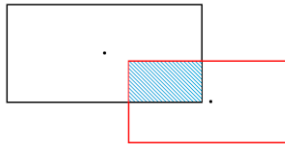
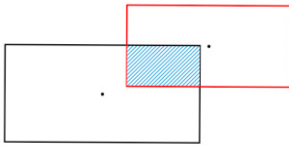
$$S_i = C_i * K_i \quad (2-1)$$

$$S_0 = \begin{bmatrix} M_i - X_0 + \frac{1}{2}(C_i + C_0) \\ * \left[N_i - Y_0 + \frac{1}{2}(K_i + K_0) \right] \end{bmatrix} \quad (1-2)$$

$$S_0 = \begin{bmatrix} M_i - X_0 + \frac{1}{2}(C_i + C_0) \\ * \left[-N_i + Y_0 + \frac{1}{2}(K_i + K_0) \right] \end{bmatrix} \quad (2-2)$$

$$R = \frac{S_0}{S_i} \quad (1-3)$$

$$R = \frac{S_0}{S_i} \quad (2-3)$$



$$\text{Case3:} \begin{cases} X_0 - M_i < 0 \\ Y_0 - N_i > 0 \end{cases}$$

$$S_i = C_i * K_i \quad (3-1)$$

$$S_0 = \begin{bmatrix} M_i - X_0 - \frac{1}{2}(C_i + C_0) \\ N_i - Y_0 + \frac{1}{2}(K_i + K_0) \end{bmatrix} \quad (3-2)$$

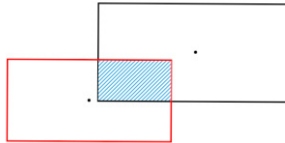
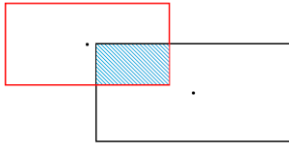
$$R = \frac{S_0}{S_i} \quad (3-3)$$

$$\text{Case5:} \begin{cases} X_0 - M_i < 0 \\ Y_0 - N_i < 0 \end{cases}$$

$$S_i = C_i * K_i \quad (4-1)$$

$$S_0 = \begin{bmatrix} -M_i + X_0 + \frac{1}{2}(C_i + C_0) \\ -N_i + Y_0 + \frac{1}{2}(K_i + K_0) \end{bmatrix} \quad (4-2)$$

$$R = \frac{S_0}{S_i} \quad (4-3)$$



When the R_1, R_2, R_3 is calculated, because the R_1 is the coincidence rate closest to the current vehicle parking space, the R_1 is defined as the central coverage rate and the R_2, R_3 is the edge coverage rate. After a lot of data investigation and calculation, we find that when the center coverage $R_1 > 50\%$, It can be judged that the parking space is almost completely occupied.

4. Result analysis

4.1. Test Model Training Results

As shown in the figure, orange is the training set and blue is the test set. It can be observed that the neural network has a good recognition effect. As can be seen from the figure, the error between the training set and the test set can basically be kept at about 0.01

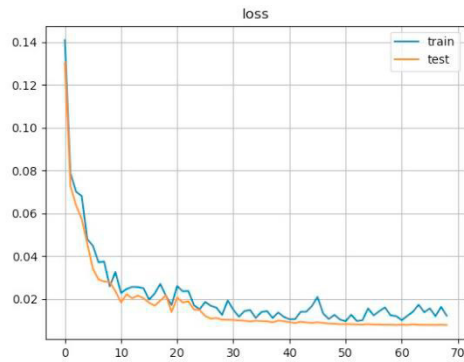


Fig.5. Loss detection of parking spaces

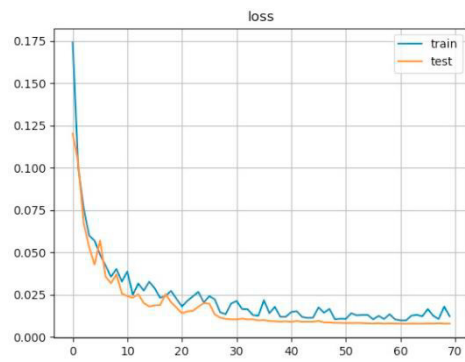


Figure.6. Vehicle detection Loss values



Fig.7. Schematic illustration of vehicle identification results

4.2. Training Results of Discriminant Model of Parking Status

Under this parameter ($R_1 > 50\%$, $R_2, R_3 < 20\%$), accuracy of parking recognition is $90.2 \pm 1.3\%$.

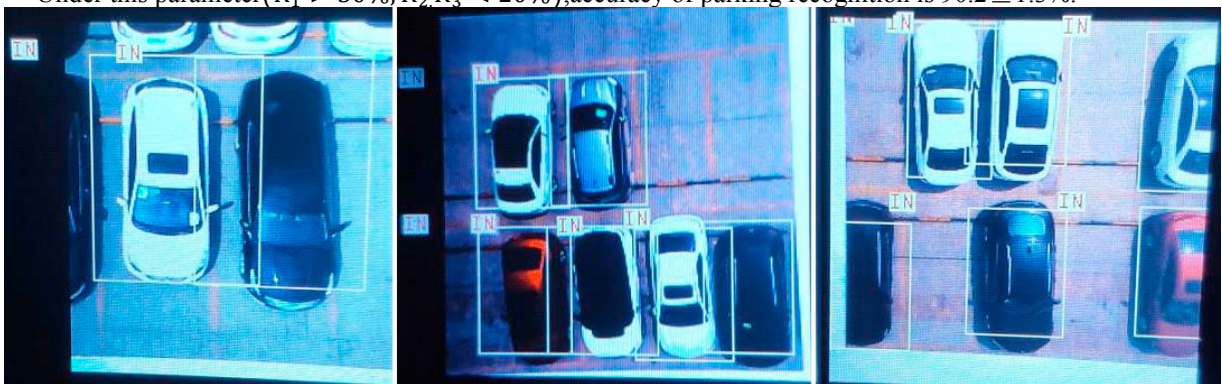


Fig.8. Schematic illustration of the discrimination of parking spaces

5. Conclusion

Based on computer vision, this paper uses YoloV2 algorithm to establish a simple and efficient parking space discrimination model. Through the learning of the algorithm, the modification of the model and the adjustment of multiple thresholds, the recognition rate is better from the actual effect, and the parking space state discrimination model is basically completed. The feasibility of this method provides a reference for the popularization of smart transportation. However, due to the limited collection of actual scenes, the accuracy of the model's decision in dealing with some special scenes needs to be improved. In the future, we will conduct multi-party field investigations to enrich the parking space scene and the movement information of the vehicle relative to the scene, so as to strengthen and improve the parking space state judgment model. In recent years, with the rapid development of the Internet, the concept of building smart cities and building smart transportation has been widely mentioned, and machine learning is its most powerful core tool. With the continuous development of artificial intelligence fields such as deep learning, smart life is just around the corner.

6. Application Outlook

- Used to construct urban intelligent traffic parking system^[6]

Based on the state judgment method of the parking space, the state information of the roadside parking space can be obtained. The mobile application can be developed so that users can obtain the query of roadside parking space in real time.

- Vehicle flow detection and information feedback on parking spaces

Indirect feedback of traffic flow and parking information is beneficial to the local management of regional vehicles by traffic management departments. At the same time, the navigation software can be linked to feedback to the traffic flow on the road and parking space information near the destination.^[7]

- Parking charges and reverse search

Setting up a camera system in a specific area, improving the algorithm and identifying the vehicle license plate can monitor the time and place of parking, and can collect parking fees in conjunction with the application.

Because the position information of the vehicle is recorded when parking, the reverse search of the vehicle can be realized by app when the user forgets the parking area.

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